HERIOT WATT UNIVERSITY

School of Mathematical and Computer Sciences

Actuarial Mathematics and Statistics

F70LA

Life Insurance Mathematics A

IN-PERSON EXAMINATION

December Examination and Assessment Diet 2024-2025

Duration: 2 Hours

Total marks: 60

Answer all 7 questions

Resources Included within Exam Paper: None
Additional Resources to be Supplied to Candidates: Orange book entitled `Formulae and Tables for Actuarial Examinations' on its spine.
Calculator Permitted: Yes

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

I understand that infringing this statement would represent an academic misconduct offence subject to disciplinary action according to the University Regulations and Policy regarding Academic Misconduct, with possible significant consequences for degree progression and final degree outcome.

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1. The force of mortality of Tyrannosaurus Rex was three times the force of mortality of Diplodocus, at all ages.

(a) Show that if ${}_tp_x^\star$ represents the survival probability of Tyrannosaurus Rex and ${}_tp_x$ represents that of Diplodocus, then

$$ {}_tp_x^\star = \left({}_tp_x\right)^3 .$$

[2 marks]

(b) Assume that Diplodocus' force of mortality was $\mu_x = 0.0125$, at all ages $x \geq 0$.

(i) State the limiting age under this mortality model.

(ii) Calculate the difference between the life expectancy from birth of Tyrannosaurus Rex and Diplodocus.

[3 marks]

[Total 5 Marks]

2. (a) For a particular select life table with a select period of 1 year,

$$\ell_{80} = 10,000, \qquad p_{79} = 0.95 \qquad \text{and} \qquad q_{[79]} = 0.6q_{79}.$$

Calculate $\ell_{[79]}$.

[3 marks]

(b) Explain why a life office uses select life tables to calculate premium rates for term assurances, and not national mortality tables.

[2 marks]

[Total 5 Marks]

3. The following table shows an extract from a life table.

x	ℓ_x
60	99,813
61	97,702
62	95,046

Calculate ${}_{1.25}p_{60}$ using the uniform distribution of deaths assumption between integer ages.

[Total 3 Marks]

4. A life insurance company offers an increasing term assurance that provides a benefit payable at the end of the year of death of £50,000 in the first year, increasing by £1,000 on each policy anniversary.

The premium basis is as follows:

Mortality: AM92 Ultimate
Interest: 6% per annum effective
Expenses: Nil

Calculate the single premium for a five year policy issued to a life aged 60 exact.

[Total 6 Marks]

5. (a) Let n be a positive integer. Describe the cashflows of the annuity which has expected present value

$$\ddot{a}_{\overline{n}|} + {}_n|\ddot{a}_x.$$

[2 marks]

(b) Without doing any calculations, explain how the expected present value would change as n increases.

[2 marks]

(c) Fix the term $n = 10$ years and age $x = 60$ years. Using the AM92 Ultimate life table and an annual effective interest rate of 4% per annum, calculate the expected present value of the annuity.

[4 marks]

(d) Write down the present value random variable of the annuity.

[2 marks]

[Total 10 Marks]

6. On 1 January 2004, an insurer issued a whole life insurance contract to a life aged 25. The sum assured £100,000 is payable immediately upon death. Premiums are payable monthly until death.

The pricing basis is as follows.

Mortality:	AM92 Select
Interest:	6% per annum effective.
Expenses:	Initial: £250 plus 120% of the first monthly premium.
	Renewal: Nil.
	Claims: £300 at the time of payment of the sum assured.

(a) Using the equivalence premium principle, calculate the monthly premium.

[10 marks]

On 31 December 2023, the policyholder requests that the policy becomes paid-up, i.e. the policyholder wishes to pay no more premiums but does not want to cancel the policy. Therefore, the contract remains a whole life insurance. However, the sum insured is reduced to reflect that no more premiums are payable.

Suppose the gross premium policy value is £10,995 on 31 December 2023, just before the policy becomes paid-up.

The following basis is used to calculate the alteration.

Mortality:	AM92 Ultimate
Interest:	6% per annum effective.
Expenses:	Renewal: Nil.
	Claims: £500 at the time of payment of the sum assured.
	Cost of the alteration: £995.

(b) Using the gross premium policy value stated above, determine the paid-up sum insured.

[5 marks]

[Total 15 Marks]

7. (a) Describe briefly three standard methods of calculating bonuses annually on a life insurance contract.

[3 marks]

Today, a man aged 45 buys a 15-year with-profits endowment insurance with a basic sum assured of £100,000. Simple reversionary bonuses of 6% per annum vest at the end of each policy year (while the insured is alive) and death benefits are payable at the end of year of death.

Premiums of amount £P per annum are payable annually until age 60 or earlier death.

The pricing basis is as follows.

Mortality:	AM92 Ultimate.
Interest:	4% per annum effective.
Expenses:	Initial: £200.
	Renewal: £30.
	Claims: Nil.
Bonus:	Simple reversionary bonus of 6% per annum vesting at the end of each policy year.

(b) Using the equivalence premium principle, calculate the annual gross premium £P.

[11 marks]

(c) Without doing any calculations, explain how the insurer's profits on the contract would be affected by the insured life dying in the first 2 years of the contract. Assume that the policy experience is as assumed in the pricing basis.

[2 marks]

[Total 16 Marks]

END OF PAPER